

PROJECT CONCEPTION & DESIGN THINKING REPORT

Detailed Conception, Persona Identification, and Problem Definition

1. Project Background & Initiation

In modern education, assessing conceptual understanding via verbal communication is key. However, manual grading of spoken explanations is time-consuming and subjective. The Voice-Based Concept Understanding Analyser (VBCUA) was conceived to automate this process locally, privately, and securely.

2. User Personas & Empathy Map Details

- **Student Persona**: Needs immediate feedback on oral explanations, worries about grading bias, and wants actionable metrics to improve speech fluency.
- **Instructor Persona**: Manages large classes, spends excessive hours listening to audio recordings, and seeks an objective helper to check semantic concept matching and speech pacing.

3. Brainstorming & Ideation Phase

During the ideation phase, various architectural approaches were evaluated. The team decided on a local-first AI system combining OpenAI Whisper for transcription, Sentence-BERT for semantic similarity matching, and Librosa for extracting acoustic characteristics (volume and silence pauses). This avoids external API costs and secures user privacy.

4. Problem-Solution Fit

The fit is achieved by mapping the candidate's spoken text directly to stored reference concepts in a local relational database, calculating a combined content-delivery score, and generating a professional, downloadable feedback report in PDF format.